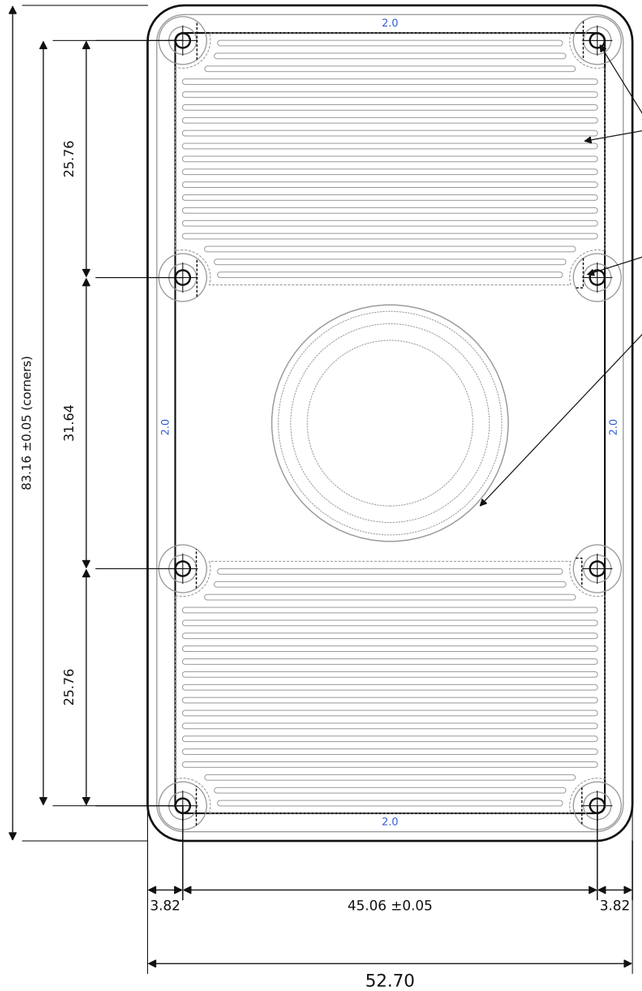
## BACK FACE (outer)  SCALE 1.5:1




| POS | Ø0.10 (M) | A | B |
|---|---|---|---|

**8×  M2 THREAD, THRU**

Ø1.6 TAP DRILL  •  TAP FROM BACK FACE

Ø3 SPOTFACE, BACK (depth per Detail B)

*C3 = hole-pattern pitch ±0.05   C4 = M2 thread*

*SECTION A-A → edge profile*  *DETAIL B → corner boss*

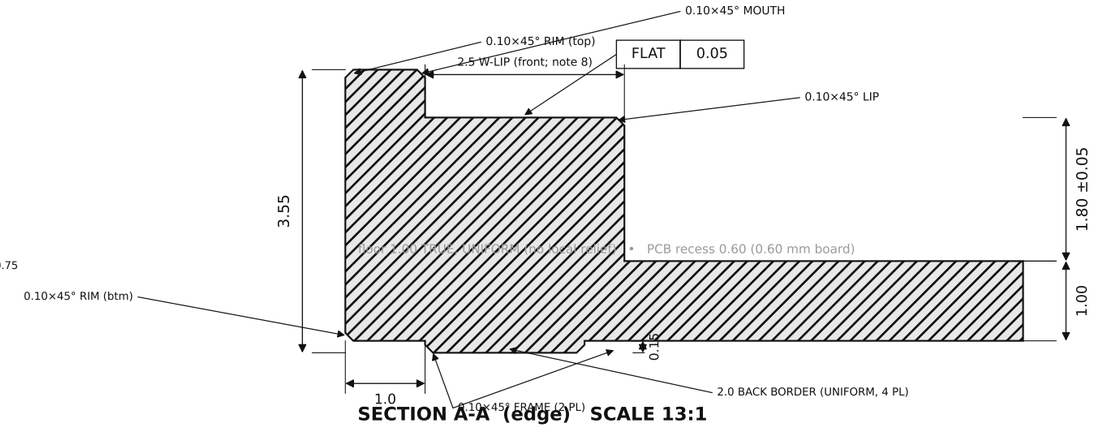
### REAR ART — MACHINED-IN GEOMETRY  (REFERENCE; STEP GOVERNS)

REEDED FIN FIELDS ×2 (PLAN, dashed zones):  19 RIBS/FIELD, 0.60 WIDE ON 1.399 PITCH; GROOVES Ø0.6 END-MILL ONLY (0.80 WIDE), VALLEY 0.60 BELOW ART FIELD (0.40 WEB); RIB TOPS +0.10 — 0.05 UNDER BEARING PLANE.

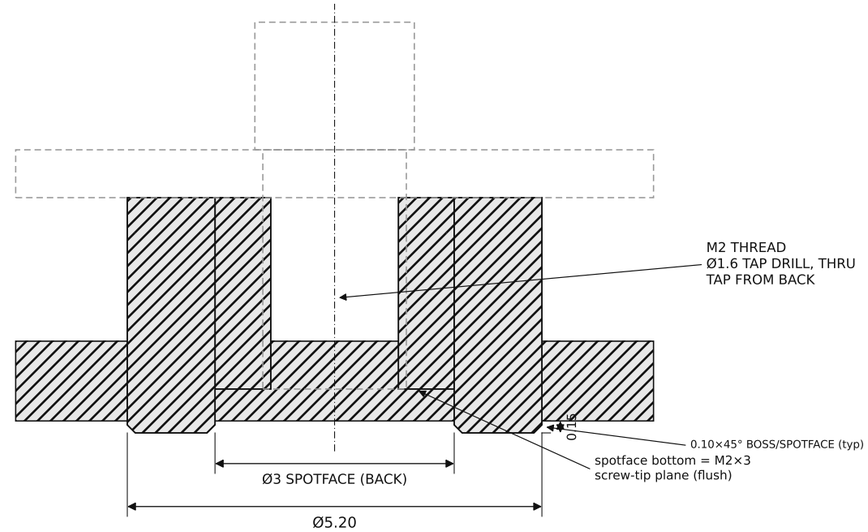
MEDALLION (PLAN, circles):  Ø25.7 COIN SUNK 0.45 BELOW ART FIELD (Ø0.4 END MILL), COUNTERS/REST 0.25 (Ø0.3 REST PASS); CRESTS (RING TEXT, RIM, HOOP, MONOGRAM, SERIAL) LEFT AT THE FRAME'S BEARING PLANE (−0.15), VERTICAL WALLS, NO DETACHED ISLAND UNDER 0.55. DO NOT EDGE-BREAK CREST EDGES. SERIAL ("No 001") IS PER-UNIT VARIABLE TEXT.

OP ORDER: ALL BACK-FACE FEATURES BEFORE CAVITY HOLLOW (SOLID-STOCK RULE). STRESS-RELIEVED STOCK — STATE CONDITION ON QUOTE.

## SECTION A-A (edge)  SCALE 13:1




*C1 = cavity depth   C2 = PCB-rest-plane flatness   (back face down; PCB drops in from top)*
*PCB + M2×3 BRASS SCREW — REF, NOT THIS PART*




## DETAIL B  (corner boss)  13:1

### CRITICAL DIMENSIONS  —  C1 & C3 TOLERANCED ±0.05 (MARKED ON VIEWS); ALL OTHER FEATURES PER ISO 2768-1 (MEDIUM)

| | | |
|---|---|---|
| **C1** | Cavity depth  (boss-top plane → cavity floor) | 1.80  ±0.05 |
| **C2** | PCB-rest plane flatness  (lip + 8 bosses) | FLAT  0.05 |
| **C3** | 8× mounting holes  (x 45.06; y rows 2.87/28.63/60.27/86.03) | ±0.05  (linear) |
| **C4** | Mounting holes — tapped, thru, from back | M2  /  Ø1.6 drill |

### NOTES

1. MATERIAL: TITANIUM Gr5 (TC4) = Ti-6Al-4V GRADE 5.  [Ti Gr5 (reference; brass or 6061 acceptable -- non-magnetic, machinable)]
2. FINISH: BEAD-BLAST MATTE, ALL OVER. NO MARKING SERVICE — THE REAR ART (REEDED FIELDS + MEDALLION) IS MACHINED-IN, IN THE STEP (SEE REAR ART BLOCK). BEARING PLANE (FRAME + MEDALLION CRESTS) HAND-LAPPED BY CUSTOMER AFTER BLAST — NO SHOP ACTION.
3. GENERAL TOLERANCE PER ISO 2768-1 (MEDIUM). C1-C4 TOLERANCED AS LISTED; C1 & C3 = ±0.05. DATUMS: A = LEFT EDGE, B = BOTTOM EDGE, C = PCB-REST PLANE.
4. BRACE REGISTRATION: THE RESIN H-BRACE REGISTERS TO THIS SHELL BY FITMENT - ITS 4 OUTBOARD RAILS + THE COMPONENT POCKETS + THE BOARD PRESS-FIT. NO LOCATOR PILLARS; THE CAVITY FLOOR IS A FULL 1.00 EVERYWHERE.
5. EDGE BREAKS - ALL 0.10x45°, FELT NOT SEEN, MODELED.  CALLED OUT ON SEC A-A / DETAIL B:  RIM = outer rim (top & bottom);  MOUTH = recess mouth (around PCB);  LIP = inner (cavity-side) lip edge;  FRAME = proud back-frame bottom edges;  BOSS = boss + spotface bottom edges (Detail B).  BREAK ALL OTHER EXPOSED EDGES 0.10x45°.  CONCAVE JUNCTION CORNERS (BOSS
6. FLOOR IS A TRUE 1.00, UNIFORM ACROSS THE CAVITY, AT THE ~1.0 Ti MIN-WALL GUIDANCE SO IT ALSO CLEARS ALUMINIUM / COPPER / STAINLESS.
   CAVITY 1.80 ±0.05 -> 1.75 WORST-CASE, MINUS WS17 1.70 MAX (DATASHEET CASE WS17 HEIGHT) = 0.05 NON-CONTACT; THE BRACE + SOLAR-CELL SANDWICHES CARRY THE BOARD.
7. NO U7 RELIEF POCKET. U7 IS THE 0.90 mm MB85RC512TY DFN-8, WELL UNDER THE 1.80 CAVITY, SO THE FLOOR IS UNIFORM 1.00. (A POCKET EXISTED FOR A 1.75 SOIC-8 THE v4 BOARD DOES NOT CARRY.)
8. FRONT SUPPORT LIP IS ASYMMETRIC (SECTION A-A): WEST 2.5 / NORTH 2.0 / SOUTH 2.0 FOR PCB RIGIDITY; EAST 1.0 CLEARING THE JP1/TP1 PADS, THE NFC COIL (A GROUNDED LIP WOULD DETUNE IT), AND C7 (x49.55, THE ONE EAST-EDGE PART LEFT AFTER v4 REMOVED THE Q1/U4/R7/R9 CLAMP CLUSTER + D9/D10/D11); WIDENING TO 2.5 ONLY AT THE SOUTH END (y0-10, CLEAR C
9. NO INTERNAL RIBS OR SUPPORT POSTS: THE RESIN BRACE (SEPARATE PART) CARRIES CENTER SUPPORT. A PCB LAYOUT CHANGE = A BRACE REPRINT, NOT A SHELL RE-MACHINE.
10. PCB RECESS = 0.60 DEEP (RECEIVES THE 0.60 mm BOARD; SLIP FIT, NOT A PRESS FIT). 3D STEP GOVERNS ALL GEOMETRY; STL NOT FOR CNC.
11. PART = BACK-SHELL ONLY. PCB, RESIN BRACE, AND 8× M2×3 BRASS SLOTTED-CHEESE SCREWS (HEAD Ø3.8) SUPPLIED SEPARATELY.

### GENERAL TOLERANCES  (UNLESS OTHERWISE NOTED, mm)

| | |
|---|---|
| LINEAR & HOLE-POSITION DIMS | **ISO 2768-m** |
| CAVITY DEPTH  C1  (SECTION A-A) | **1.80 ±0.05** |
| HOLE-PATTERN PITCH  C3  (8 HOLES) | **x 45.06 / y 25.76-31.64-25.76  ±0.05** |
| PCB-REST FLATNESS  C2 | **FLAT 0.05** |

### SOLAR-GLOW DRH v4.0  —  Ti BACK-SHELL (0.6mm-BOARD DUMB BOX)

| | |
|---|---|
| DWG  solar-glow-drh-v4_0-backshell-0p6b-brace | REV  v4.0 |
| MATERIAL  Ti Gr5 (TC4) | FINISH  BEAD-BLAST + LAP (CUSTOMER) |
| UNITS  mm | SCALE  AS NOTED |
| BBOX 52.70 x 90.80 x 3.55  3rd-angle | SHEET 1/1 |